

Casey E. Middleton, PhD

caseymiddleton67@gmail.com • [LinkedIn](#) • [Personal Website](#)

EDUCATION

University of Colorado, Boulder, CO USA

PhD, Computer Science (May 2025)

Advised by Dr. Daniel Larremore

Multiscale Infectious Disease Dynamics: Linking Epidemiology and Testing for Outbreak Neutralization

Rhodes College, Memphis, TN USA

B.S. [Biomathematics](#) — *cum laude*

RESEARCH PUBLICATIONS

Middleton, C., Eales, O., McCaw, J., and Shearer, F. Incorporating vaccine effects into epidemiological models: common pitfalls and solutions. [under review]

Middleton, C. and Larremore, D. Statistical Methods for Estimating the Protective Effects of Immune Markers Using Test-Negative Designs. *American Journal of Epidemiology* (2025). doi:[10.1093/aje/kwaf280](https://doi.org/10.1093/aje/kwaf280).

Bubar, K., **Middleton, C.**, Larremore, D., and Gostic, K. A fundamental limit to the effectiveness of traveller screening with molecular tests. *Epidemiology & Infection* **153**, e95 (2025). doi:[10.1017/S0950268825100381](https://doi.org/10.1017/S0950268825100381).

Middleton, C. and Larremore, D. Modeling the transmission mitigation impact of testing for infectious diseases. *Science Advances* **10**, 5108 (2024). doi:[10.1126/sciadv.adk5108](https://doi.org/10.1126/sciadv.adk5108).

Featured in [CU Today](#), [HealthDay News](#), [US News & World Report](#), etc.

Wu, S., Bjerke, J., **Middleton, C.**, et al. Longitudinal analysis of viral and host RNA in blood and saliva during controlled human dengue virus 3 infections. *Journal of Virology* (2026). doi:[10.1128/jvi.00373-26](https://doi.org/10.1128/jvi.00373-26).

Bubar, K.*, **Middleton, C.***, Bjorkman, K., Parker, R., Larremore, D. SARS-CoV-2 Transmission and Impacts of Unvaccinated-Only Screening in Populations of Mixed Vaccination Status. *Nature Communications* **13**, 2777 (2022). doi:[10.1038/s41467-022-30144-7](https://doi.org/10.1038/s41467-022-30144-7). Featured in [CU Boulder Today](#).

Scott, S., **Middleton, C.**, and Bodine, E.N. An agent-based model of the spatial distribution and density of the Santa Cruz Island Fox: the effects of Golden Eagle predation and Island Fox recovery. *Integrated Population Biology & Modeling*, Vol. 40 of *Handbook of Statistics*, Elsevier, 2019. doi:[10.1016/bs.host.2018.10.001](https://doi.org/10.1016/bs.host.2018.10.001)

Middleton, C., Deery, E., and Bodine, E. The Potential Impact of Using Vaccination and Insect Repellent to Control the Spread of Yellow Fever. *SPORA: A Journal of Biomathematics* **4(1)**: 15 – 24. doi:[10.30707/SPORA4.1Middleton](https://doi.org/10.30707/SPORA4.1Middleton). Recipient of Outstanding Undergraduate Research award.

* Authors contributed equally

AWARDS

2025 **Departmental Outstanding Research Paper Award**
Department of Computer Science, University of Colorado Boulder, Boulder, CO

2025 **3 Minute Thesis Finalist** (science communication)
University of Colorado Boulder, Boulder, CO

2022 **Departmental Outstanding Research Award**
Department of Computer Science, University of Colorado Boulder, Boulder, CO

2020 **Olke C. Uhlenbeck Graduate Fellowship**
BioFrontiers Institute, University of Colorado Boulder, Boulder, CO

2018 **Outstanding Senior in Mathematics**
Rhodes College, Memphis, TN

2017 **Outstanding Undergraduate Research in Biomathematics and Ecology Scholarship**
Intercollegiate Biomathematics Alliance

Robert Allen Scott Award in Mathematics
Rhodes College, Memphis, TN

SKILLS

Programming & Computation

- *Basic Programming Languages*: Python, R, C⁺⁺, SQL
- *Modeling & Computation Software*: MATLAB, NetLogo, Stan, L^AT_EX
- *Geographic Information Systems*: ArcGIS, QGIS

Constructing & Analyzing Mathematical Models

- **Mechanistic**: ordinary differential equations, network, geospatial, individual-based models of population-level & within-host dynamics
- **Statistical modeling & inference**: Bayesian and frequentist approaches to estimation and prediction, including MCMC, forecasting, regression-based nowcasting, and correlates-of-protection analyses

Scientific Communication

- Passionate about translating scientific findings into policy recommendations
- Storytelling with data visualizations, including uncertainty
- Scoping presentations to be appropriate for intended audience

Certificates

- Research Communication & Presentation microcredential, University of Colorado Boulder
- Interdisciplinary Quantitative Biology Research certificate
- SISIMID certificates: Infectious Diseases, Immunology and Within Host Models; MCMC 1/2; Machine Learning for Infectious Disease Modeling

PROFESSIONAL EXPERIENCE

2025 – Current	Research contractor University of Melbourne Modeling heterogeneous vaccine effects; examining sources of bias in effect estimation
2024	Summer ORISE Fellow Center for Forecasting and Outbreak Analytics (CFA) at CDC Developing performance analysis pipeline for SARS-CoV-2 variant nowcasting methods
2020 – 2025	Graduate Research Assistant Department of Computer Science, University of Colorado Boulder Focus on the utility of diagnostics and vaccination to curtail disease transmission
2019 – 2020	Contract Data Analyst Data analysis, visualization, & policy recommendations for City of Memphis Lead Teacher KIPP Memphis Collegiate High School Tenth grade geometry teacher
2018 – 2019	Americorps: Volunteer in Service to America Neighborhood Preservation, Inc. Data analytics for housing policy recommendations
2016 – 2018	Teaching Assistant Department of Mathematics and Computer Science, Rhodes College Classes: <i>Mathematical Modeling with Biological Applications</i> , <i>Applied Calculus</i> Senior Administrator for Applied Mathematics Tutoring Department of Mathematics and Computer Science, Rhodes College
2016	Biomathematics Research Fellow Department of Mathematics and Computer Science, Rhodes College Generating social structure algorithms for yellow-bellied marmot populations
2015 – 2016	Calculus Tutor Department of Mathematics and Computer Science, Rhodes College

RESEARCH PRESENTATIONS

† : ORAL PRESENTATION

- “Statistical Methods for Estimating the Protecting Effects of Immune Markers Using TNDs” †
EPIDEMICS, San Diego, CA, December 2025.
- “Testing, Testing, 1, 2, 3.” †
3 Minute Thesis, Boulder, Colorado, February 2025.
- “Fundamental Limit to the Effectiveness of Traveler Screening.”
EPIDEMICS 2023, Bologna, Italy, November 2023.
MIDAS 2023, Atlanta, GA, September 2023. †
- “How Should We Test for Infectious Diseases?” †
Computer Science Departmental Colloquium, University of Colorado Boulder, February 2023.
- “A Generalized Modeling Framework for the Mitigation Potential of Testing.” †
MIDAS 2022, Bethesda, Maryland, September 2022.
- “Modeling the Impact of Unvaccinated-Only Screening in Populations with Mixed Vaccination Status.”
Ecology and Evolution of Infectious Diseases (EEID) 2022, Atlanta, Georgia, June 2022.
- “A Probabilistic Model of Disease Mitigation via Testing.” †
Math Bio SuperGroup, Boulder, Colorado, November 2021 (virtual).
- “Forecasting Spring Green-Up in Deciduous Forests.” †
Quantitative Exploration SuperGroup, Boulder, Colorado, May 2021 (virtual).
- “Predicting the Impact of Community Testing on SARS-CoV-2 Transmission.” †
Biofrontiers Institute, Boulder, Colorado, March 2021 (virtual).
- “Predicting the Impact of Community Testing on SARS-CoV-2 Transmission.” †
Biofrontiers Institute, Boulder, Colorado, December 2020 (virtual).
- “Modeling Soil Moisture Impacts on Perennial and Annual Crop Production.” †
Biofrontiers Institute, Boulder, Colorado, October 2020 (virtual).
- “Reducing False Alarm Occurrence in the City of Memphis.” †
Mayor’s Dashboard, Memphis, TN, May 2019 (virtual).
- “An Accountability System for Rental Property Management Companies.” †
Department of Public Works at City Hall, Memphis, TN, April 2019.
- “Urban Mathematics: Using Applied Mathematics in Neighborhood Revitalization.” †
Blight Elimination Steering Team Meeting, Memphis, TN, March 2019.
- “The Potential Impact of Using Vaccination and Insect Repellent to Control the Spread of Yellow Fever.”
Intl. Symposium on Biomathematics and Ecology Education and Research, Normal, IL, October 2017.

SERVICE

Service to Infectious Disease Modeling Community

- **SISMID Advisory Board Member** (2024 & 2025): Providing content guidance for Summer Institute in Statistics and Modeling in Infectious Diseases

Service to University of Colorado, Boulder

- **Graduation Speaker** (2025): Selected to address 4000 attendees at the CS graduation ceremony
- **Quantitative Exploration and Discussion** (2021 - 2024): Organizing monthly workshops for the Biofrontiers Institute with a focus on computational skills and quantitative biology
- **Women in Mathematics** (2022): Calculus tutor

Service to Rhodes College Department of Mathematics and Computer Science

- **Computational writing editor** (2017 - 2018): Grader and editor for continuous mathematical modeling, a writing intensive APPM course
- **Applied calculus training** (2016 - 2018): Organized tutorial sessions, training tutors to provide guidance on group projects or homework problems
- **Calculus sequence redesign** (2018): Served on a committee to redesign the calculus sequence, planning coursework to be performed in a guided lab outside of class meeting times
- **Applied mathematics tutorials** (2017 - 2018): Worked with multiple professors to design a weekly tutorial meeting for students enrolled in mathematical modeling with biological applications